INVERTIS UNIVERSITY, BAREILLY

FACULTY OF ENGINEERING & TECHNOLOGY

OFFICIAL PRACTICAL & LABORATORY MANUAL â¢ DEPARTMENT OF CSE

Laboratory Course: [BAS151] Physics Lab

Programme: Bachelor of Technology in Computer Science & Engineering **Manual Type: LAB MANUAL**

Prescribed w.e.f Academic Session 2023-2024 **Lab Coordinator: Department of Applied Scie**

Official Laboratory Manual: Engineering Physics & Optics

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1. Laboratory Guidelines & Safety Protocol

- Safety: Never look directly into laser beams; discharge high voltage capacitors carefully.
- Apparatus: Spectrometer, Sodium vapour lamp, He-Ne Laser, Polarimeter, Hall effect setup.
- Evaluation: Observation table accuracy (30%), least count measurement (20%), viva (20%), calculations (30%).

2. Prescribed Experiments

- Exp 1-4: Newton’s ring wavelength determination, Fresnel’s biprism, nodal slide, cane sugar polarimeter.
- Exp 5-7: Transmission grating spectral lines, laser diffraction wavelength measurement.
- Exp 8-12: Carey Foster’s bridge specific resistance, Stefan’s law, Hall coefficient and carrier density.